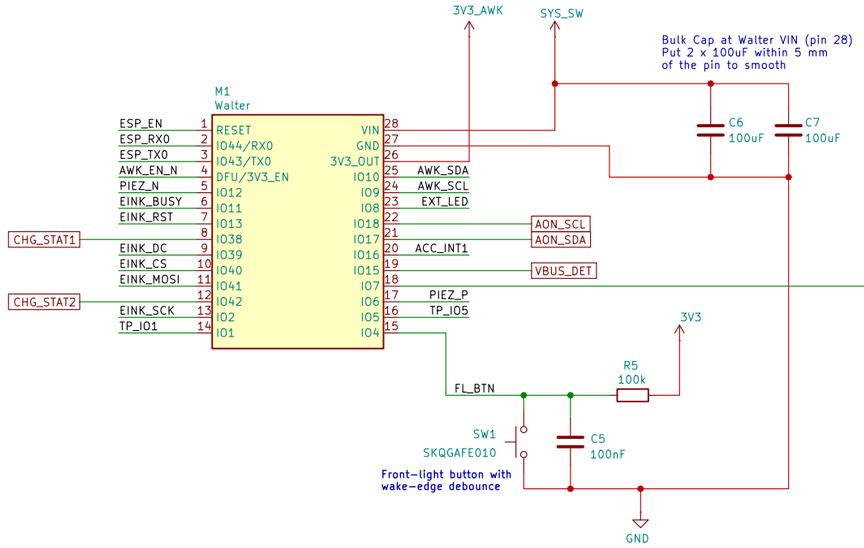
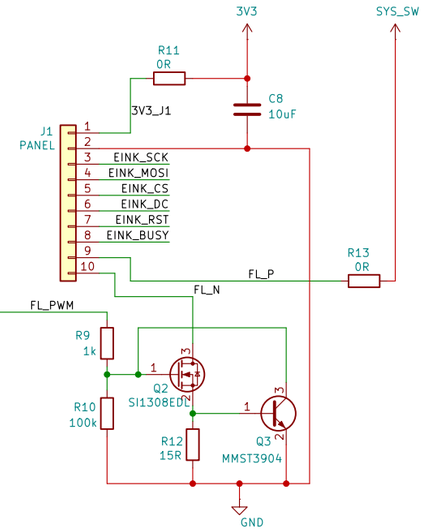
Walter ESP32 connections as represetned by 2x14 header.
The module is mounted header-side-up under the carrier.

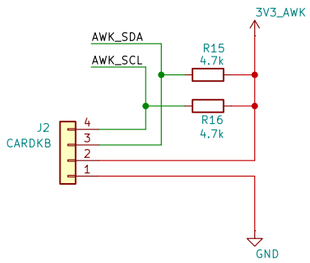


E-ink display. J3 10-pin JST SH cable connecting carrier to panel board.

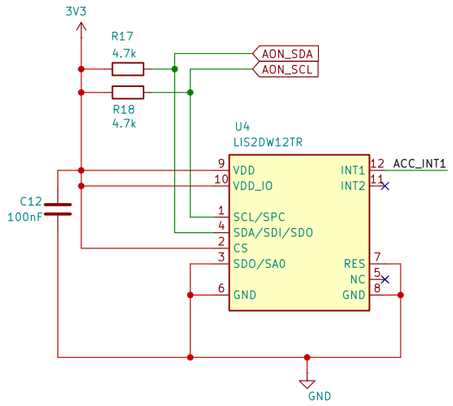
The SSD1680 ZIF, boost/charge-pump cluster, and the front-light 6-pin ZIFs live on a separate panel board. This connects the SPI channel, provides measurement points, and also provides a clamped power for the Front Light. Front Light expects 3 or 4 LEDs from Vf 3.3V to vF 2.8V.



I2C Keyboard (CardKB) header. J4 (Grove order):
1 AWK_SCL, 2 AWK_SDA, 3 3V3_AWK, 4 GND
Pull-ups to 3V3_SW so whole unit can be depowered.



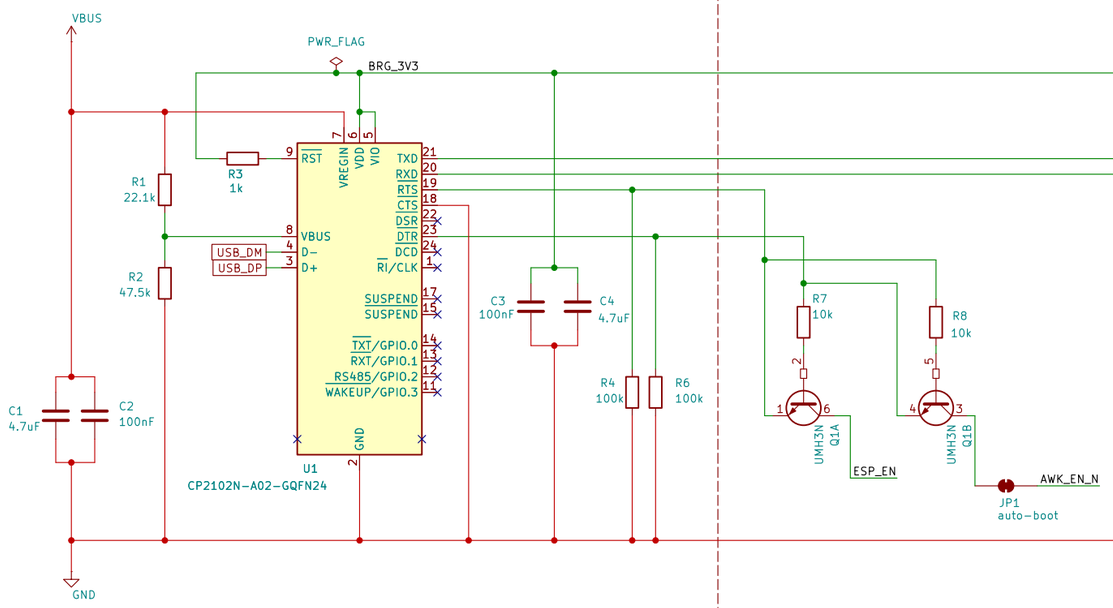
Accelerometer
U7 LIS2DW12 on I2C1 (ACC_*), SA0 low = 0x18
Always on from 3V3
CS high = I2C mode, INT1 -> IO16 (RTC wake)



Place U7 away from J7 and the panel cable; X axis along the tin's long edge; mark orientation on silk.

USB Serial connection

Bus-powered: VREGIN from VBUS
VBUS sense pin via 22.1k/47.5k
VDD = internal 3.3 V regulator output
VIO tied to VDD, 4.7uF + 100nF per power pin.
Program the USB product string with
Simplicity Studio XpressConfigurator after assembly.

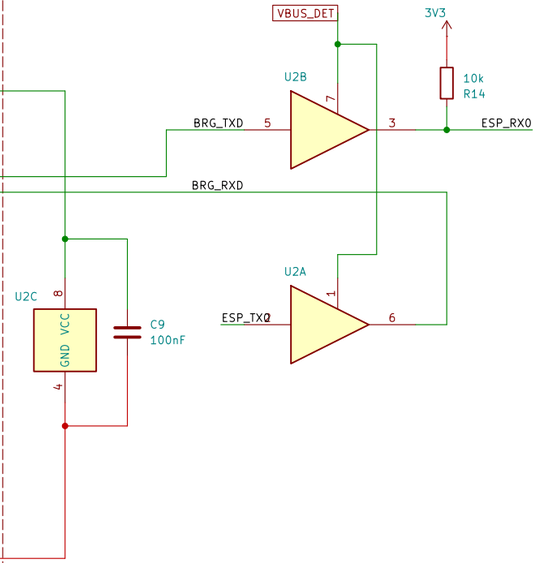


Auto-reset (Espressif):
Q2 DTR->EN, Q3 RTS->IO0.
JP1 (closed) in Q3's collector lets
auto-boot be disabled reset still works.

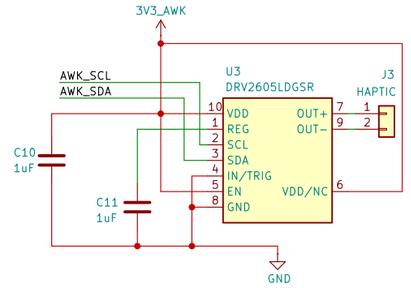
Q2/Q3 are the two halves of one UMH3N (Q2A/Q2B)
The Walter already pulls EN up (10k)
IO0 has only the ESP32 internal pull-up.

Buffer to disable the serial when no VBUS

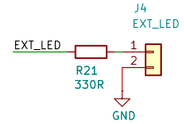
U6 SN74LVC2G126: both OE from VBUS_DET.
A: ESP_TX0 -> BRG_RXD. B: BRG_TXD -> ESP_RX0.
10k keeps ESP_RX0 high with no cable.
U6 VCC is BRG_3V3 (bus-powered).
with no cable the buffer is unpowered (Ioff outputs) so
no LVC quiescent/float current on the battery.



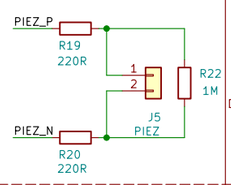
Haptic (DNP): DRV2605L on I2C1 (0x5A)
OUT+/- to J7



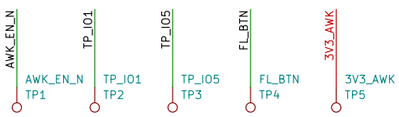
External LED Header



External Piezzo Header



Test pads

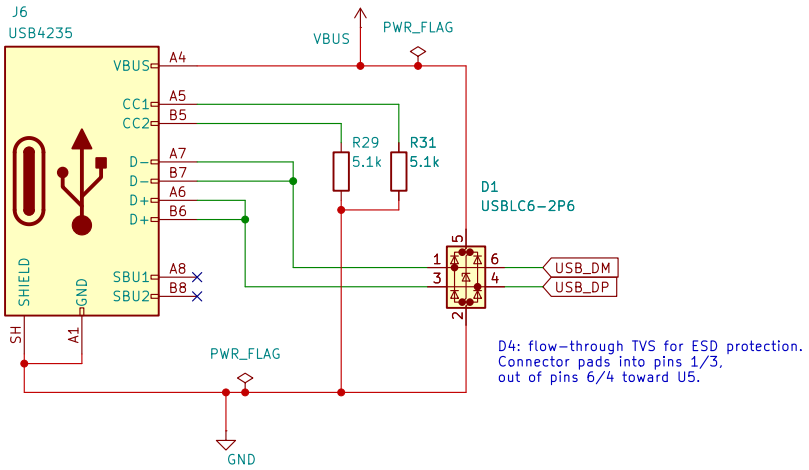


Sheet: /Core/
File: core_components.kicad_sch

Title:

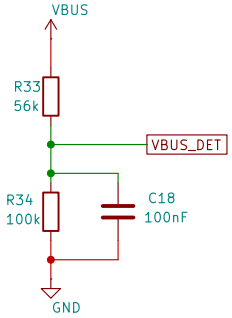
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 1/2

USB-C: vertical GCT USB4235-03-C (J1), 6.5 mm tall. CC1/CC2 5.1k to GND (UFP).
D+/D- A and B pads joined at the connector, through the TVS D4 to the bridge U5.

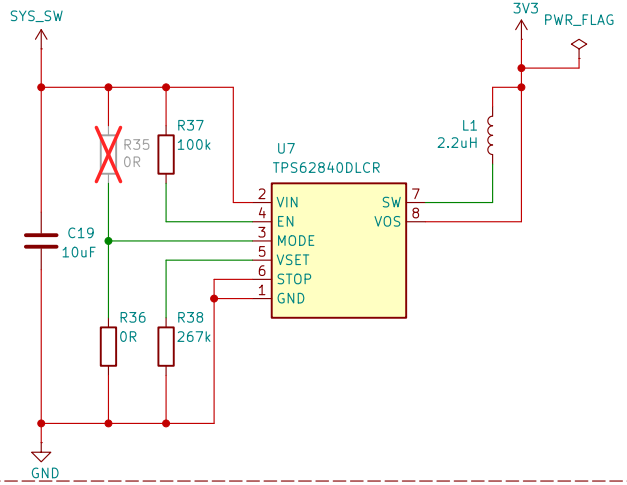


D4: flow-through TVS for ESD protection.
Connector pads into pins 1/3,
out of pins 6/4 toward U5.

VBUS detect; enabls U6.
56k/100k divider for
5V -> 3.2V level conversion



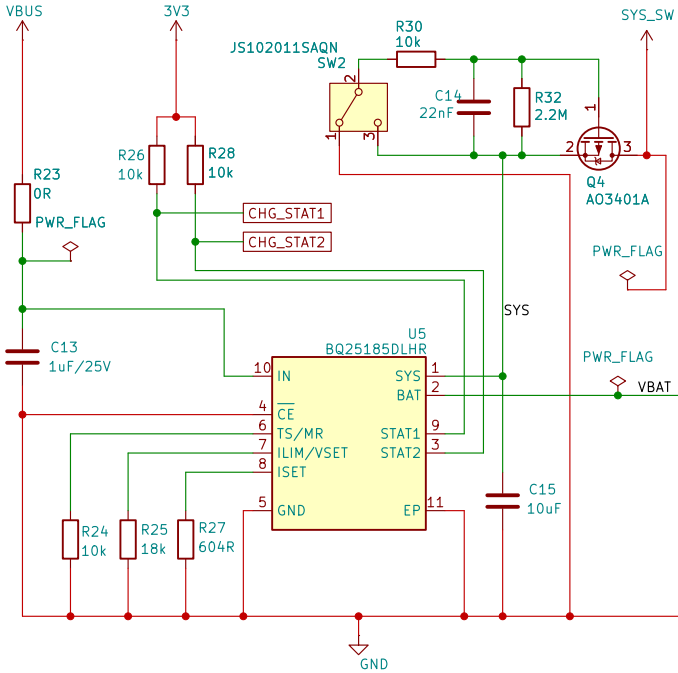
3.3 V always-on rail: TPS62840DLCR buck from SYS_SW, 2.2uH, 10uF in/out.
VSET 267k = 3.3 V.
MODE low = power-save (DNP 0R to 3V3 can force PWM)



Charger and on/off switch.

ISET 604R = 497 mA fast charge.
ILIM/VSET 18k = 500 mA input limit, 4.20 V

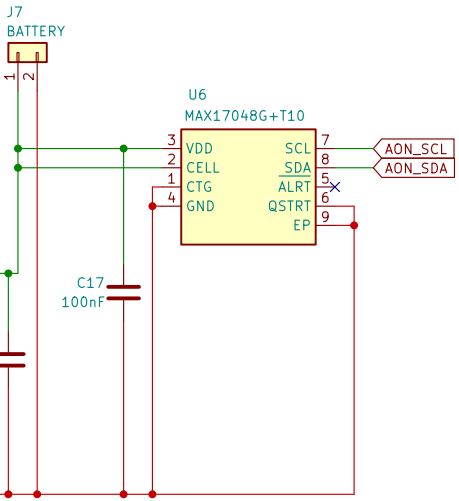
Q2 2.2M gate pull-up. (1.9 uA when off)
10k + 22nF soft-start limits inrush to Walter Bulk Caps.



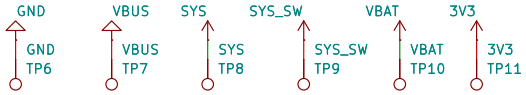
Battery + Fuel Gauge

Battery: J2 (pin 1 = +)

Fuel gauge U10 MAX17048.
Before the switch to track while off, ~3 uA
I2C1 (ACC_*), addr 0x36. QSTRT and CTG low.



Test Points



Sheet: /Power/
File: power_components.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 2/2